Nibraas Khan

+1 615-756-5508 | nnibraas@gmail.com
nibraaskhan.com | in/nibraas | /nibraaska
Cambridge, MA - USA

Highlights

Research scientist with 25+ publications, shipping interpretable ML from prototype to production across healthcare, industrial sensing, and multi-agent systems. Methods span explainable AI, few-shot learning, and multimodal sensor fusion, with recent work in LLMs, VLMs, and agentic pipelines.

Education

- **Vanderbilt University** — *Aug 2020 - June 2026 Expected*
  Ph.D. Computer Science — Nashville, TN
- **Middle Tennessee State University** — *Aug 2017 - May 2020*
  B.S. Computer Science — Murfreesboro, TN

Experience

- **Mitsubishi Electric Research Laboratories** — *Jan 2026 – Present*
  Research Scientist Intern - Few-Shot Learning — Cambridge, MA
  - Developing a neural process framework for industrial sensor anomaly detection; using reinforcement learning to select optimal context set sizes under domain shift.
  - Integrating entropic partial optimal transport with Bayesian network structure for cross-domain feature alignment and pseudo-sample generation; improving R^2 and reducing training time.
  - Preparing two Python libraries for industrial deployment and publication: one for the neural process pipeline and one for optimal transport, each with evaluation harnesses, baselines, and ablation studies.
  - Collaborating on research applying VLMs and VLAs to robotic manipulation and foundation models for automated scientific discovery in industrial sensing.

- **Actual Insight** — *Oct 2025 - Jan 2026*
  Applied Scientist Intern - Multi-Agent Systems — Nashville, TN
  - Designed and deployed multi-agent LLM pipelines that extracted actionable insights from enterprise client data; implemented provenance tracking so each output linked back to specific source material.
  - Built privacy-preserving guardrails across the multi-agent pipeline; enforced role-based data scoping so each agent accessed only authorized client segments, with PII redaction and audit logging.
  - Scoped natural language to SQL for small and medium-sized business data using a multi-agent workflow with LangGraph; used locality-sensitive hashing with caching to inject only relevant schema context.

- **Vanderbilt University - Robotics and Autonomous Systems Lab** — *May 2020 - Present*
  Graduate Researcher — Nashville, TN
  - **Tiered eXplainable AI (xAI) Framework**
    - Shipping a Python xAI library that discovers motion primitives from time-series, aligns them to prototypes, and produces tiered explanations for stakeholder review.
    - Automating developer and stakeholder views with saliency over raw signals, primitives visualizations, prototypes, and counterfactual exploration for model audits.
    - Providing model-agnostic APIs that integrate with PyTorch and TensorFlow and fit standard MLOps workflows; delivering clear documentation and examples.
  - **Real-Time Precursor Detection System (REACT)**
    - Training few-shot PyTorch model in low-data clinical environment; engineering a multimodal feature pipeline from wearable time-series and audio into 50+ features; tuning with Optuna to reach 84.6% F1.
    - Operating MLOps stack on AWS EC2 with Kafka ingestion; using Great Expectations for data quality; orchestrating weekly retraining in Airflow trains in Docker and deploys artifacts for to an iOS fleet.
    - Shipping an iOS app for data collection, real-time inference, and alerts (Apple Watch); using xAI library to surface prototype matches and counterfactuals; logging outcomes for continuous model monitoring.
  - **End-to-End Wearable System for Biomechanical Analysis**

  * Building the full stack from hardware to firmware to cloud to app; developing a custom PCB with IMUs/ESP32, streaming firmware, ingestion and services, and a React app with a 3D avatar.
  * Leading data collection and real-time analysis on AWS; streaming 100 Hz from 10 IMUs to the react app and AWS; delivering analysis to feedback in <1s for athlete guidance.
  * Using the xAI library to explain movements as motion primitives; comparing against expert movement with a Siamese model; driving actionable feedback for training sessions for any sport or activity.
- **Deep Learning Pipeline for Digital Biomarker Discovery**
  * Training a TensorFlow/Keras Bi-GRU with 4-head attention to model activity sequences and identify biomarkers differentiating MCI vs healthy for clinical insights; tuning with Optuna; reporting 86% F1.
  * Architecting pipelines for time-series analysis; leading a coding team for data labeling; fixing unsynced IMU starts, misaligned timestamps, and missing labels; delivering reproducible datasets and code.
  * Applying the xAI library to decompose sequences into motion primitives and generate explanations.
- **Mentorship & Guidance**
  * Scaled a cross-institution mentorship program; mentored 30+ students (2 to PhD, 3 to master's); built onboarding docs, starter repos, and tests that improved reproducibility and delivery velocity.
  * Led cross-functional university teams to deliver tools, managing agile cycles and technical mentorship across hardware, firmware, and mobile development.

VENTURE CREATION

- **JumpStart** *Jan 2023 – Present*
  Co-Founder Nashville, TN
  - Co-founded a community program where students gain career-ready skills by building pro-bono software for real clients, supported by technical workshops and industry mentorship.
  - Guided students to ship real-time ML systems, including a UAV ground-control station (YOLOv8 + DeepSORT + WebRTC) and an XGBoost clinical analysis tool with online learning for Vanderbilt's TRIAD.

SELECTED PUBLICATIONS

- **Khan, N.**, Cole, K., Sarkar, N. 2026. *Toward Assessing Functional Decline in Mild Cognitive Impairment Using Wearable Sensors and Explainable Machine Learning*. Digital Health.
- **Khan, N.**, Staubitz, J., Plunk, A., Shragge, I., Brooks, J., Wright, S., Brewer, A., Bozkurt, A., Weitlauf, A., Sarkar, N. 2026. *Early Warning of Behavioral Escalation in Autistic Children Using Machine Learning and Wearable Sensors*. IMWUT.
- **Khan, N.**, Plunk, A., Zhaobo, Z., Adiani, D., Staubitz, J., Weitlauf, A., Sarkar, N. 2024. *Pilot Study of a Real-time Early Agitation Capture Technology (REACT) for Children with Intellectual and Developmental Disabilities.* Digital Health. https://doi.org/10.1177/20552076241287884
- Ghosh, R., **Khan, N.**, Migovich, M., Tate, J., Maxwell, C., Latshaw, E., Newhouse, P., Scharre, D., Tan, A., Colopietro, K., Mion, L., Sarkar, N. 2024. *User-Centered Design of Socially Assistive Robotic Combined with Non-Immersive Virtual Reality-based Dyadic Activities for Older Adults Residing in Long Term Care Facilities.* arXiv. https://doi.org/10.48550/arXiv.2410.21197
- Wang, H., **Khan, N.**, Chen, A., Sarkar, N., Wisniewski, P., Ma, M. 2024. *MicroXercise: A Micro-Level Comparative and Explainable System for Remote Physical Therapy.* CHASE. https://doi.org/10.1109/CHASE60773.2024.00017

SKILLS

- **Languages:** Python, Typescript, C#, Kotlin, SQL, Bash
- **ML Frameworks & Data Science:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Hugging Face Transformers, OpenCV
- **Cloud & MLOps:** AWS (EC2, S3, SageMaker), Firebase, GCP, Docker, Git, GitHub Actions, Git LFS
- **Data Engineering & Orchestration:** Kafka, Airflow, Great Expectations, N8N
- **Databases:** MongoDB, SQLite, Supabase, Firebase
- **ML Specializations:** Explainable AI (xAI), Reinforcement Learning (Hierarchical RL, Policy Transfer), Deep Learning (LSTMs, GANs, Siamese Networks), Computer Vision (Object Detection & Tracking), Time-Series Analysis (Wearable Sensor Data), LLM Fine-Tuning, RAG, embeddings, prompt engineering
- **Developer Platforms & Tools:** Unity, React, React Native, Android/iOS Development, Core ML, TensorFlow Lite, Flask, WebSockets, Pytest

ADDITIONAL INFORMATION

Languages: Fluent in English, Conversational in Hindi and Urdu
Interests: Cooking, Boxing, Exploring restaurants